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25CYC222 – Data Communication and Networks Laboratory

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AIM

To install Cisco Packet Tracer software on the system and verify its successful setup for performing networking simulations.

PROCEDURE**Step 1:** Visit the Cisco NetAcad Website

- Open any web browser (Google Chrome, Mozilla Firefox, or Microsoft Edge).
- Go to the URL: <https://www.netacad.com>
- Click on the 'Sign In' button at the top right corner of the page.
- If you do not have an account, click 'Sign Up' and fill in the required details (name, email, password).
- Verify your email by clicking the link sent to your inbox, then log in to the Cisco Networking Academy portal.

Step 2: Enroll in the Packet Tracer Course

- After logging in, click on 'Catalog' in the top navigation bar.
- Search for 'Cisco Packet Tracer' in the search box.
- Select the free self-paced course titled 'Getting Started with Cisco Packet Tracer'.
- Click 'Get Started' or 'Enroll' to register for the free course — this gives you access to download the installer.

Step 3: Download the Packet Tracer Installer

- After enrolling, navigate to the Resources section of the course.
- Select your operating system: Windows (64-bit recommended), Mac, or Linux.
- Click the Download button. The file will be saved as a .exe file (Windows) or equivalent.
- Note the download location (usually the Downloads folder).

Step 4: Run the Installer

- Navigate to the Downloads folder and locate the Packet Tracer installer file.
- Double-click the .exe file to launch the Setup Wizard.
- If a User Account Control (UAC) prompt appears asking 'Do you want to allow this app to make changes?', click Yes.

Step 5: Complete the Setup Wizard

- The Cisco Packet Tracer Setup Wizard opens. Click Next to begin.
- Read the License Agreement carefully. Select 'I accept the agreement' and click Next.
- Choose the Destination Folder where Packet Tracer will be installed (default is recommended: C:\Program Files\Cisco Packet Tracer).
- Click Next, then click Install. The installation progress bar will appear.
- Wait for the installation to complete (approximately 1–3 minutes).
- When done, click Finish to close the wizard. Leave the 'Launch Cisco Packet Tracer' checkbox ticked.

Step 6: Launch and Verify Packet Tracer

- Cisco Packet Tracer opens automatically. A login window appears.

- Enter your Cisco NetAcad credentials (email and password) and click Log In.
- The main Packet Tracer workspace opens, showing: the Menu Bar at the top, the Device Category Panel at the bottom left, the Workspace (topology area) in the center, and the Realtime / Simulation Mode toggle at the bottom right.
- Verify all panels are visible and the software responds correctly by dragging a device (e.g., a PC) onto the workspace.

Step 7: Record Observations

Step	Action	Observation
1	NetAcad Account Created	Account creation successful / login verified
2	Installer Downloaded	Setup file saved to Downloads folder
3	Installation Completed	No errors during setup; Finish button appeared
4	Software Launched	Packet Tracer workspace opened successfully
5	Interface Verified	Menu bar, device panel, and workspace visible

RESULT

Cisco Packet Tracer was successfully installed and launched. The software is ready for performing networking experiments.

AIM

To identify the basic components of a data communication system and understand the function of each component using Cisco Packet Tracer.

PROCEDURE**Step 1:** Launch Cisco Packet Tracer

- Double-click the Cisco Packet Tracer icon on the desktop or search it from the Start Menu.
- Log in with your Cisco NetAcad credentials when prompted.
- The workspace opens with the topology area in the center, the device category panel at the bottom left, and the toolbar at the top.

Step 2: Identify the Device Category Panel

- Look at the bottom-left of the Packet Tracer window. You will see a row of icons representing device categories.
- The main categories are: End Devices (computers, printers), Network Devices (switches, routers, hubs), Connections (cable types), Security, WAN Emulation, and Custom Made Devices.
- Click on each category icon to see the individual devices available in that category.

Step 3: Place End Devices onto the Workspace

- Click on End Devices in the device category panel (icon looks like a monitor).
- From the sub-panel that appears below, drag and drop each of the following devices onto the workspace: PC (generic), Laptop, Server, Printer.
- Arrange them in a row in the upper half of the workspace.
- Click on each device icon to see its name label appear below it.

Step 4: Place Network Devices onto the Workspace

- Click on Network Devices in the device category panel.
- Drag and drop the following devices onto the workspace: Switch (2960), Hub (Hub-PT), Router (2911), Wireless Router (Linksys WRT300N), Access Point (AP-PT).
- Arrange them below the end devices.

Step 5: Place WAN / Other Devices

- Click on the WAN Emulation or Network Devices category and drag a Cloud device onto the workspace to represent the Internet or external WAN.

Step 6: Inspect Each Device

- Single-click on any device to see its label. Hover over it to see a tooltip with its name.
- Double-click on a device (e.g., a Router) to open its configuration window. Explore the Physical, Config, and CLI tabs to understand the device's capabilities.
- Double-click on the PC. Go to the Desktop tab and observe available applications: IP Configuration, Command Prompt, Web Browser, etc.
- Double-click on the Server. Go to the Services tab and observe the available services: HTTP, DNS, DHCP, EMAIL, FTP, etc.

Step 7: Identify and Record Functions

Record the category, purpose, and function of each identified component in the observation table below.

S.No	Component	Category	Function / Purpose
1	PC	End Device	Sends and receives data; acts as a client. Supports IP configuration, web browsing, command prompt.
2	Laptop	End Device	Portable end device; connects to wired or wireless networks; same client capabilities as PC.
3	Server	End Device	Provides network services such as HTTP (web), DNS, DHCP, EMAIL, and FTP to client devices.
4	Printer	End Device	Network-connected output device; receives print jobs from client PCs over the network.
5	Switch (2960)	Network Device	Connects multiple devices in a LAN; forwards frames using MAC address table; operates at Layer 2.
6	Hub (Hub-PT)	Network Device	Connects devices in a LAN but broadcasts all data to every port; no intelligence; Layer 1 device.
7	Router (2911)	Network Device	Connects different IP networks; routes packets using IP addresses; operates at Layer 3.
8	Wireless Router	Network Device	Combines router and Wi-Fi access point; provides internet and wireless LAN connectivity.
9	Access Point	Network Device	Extends wireless coverage in a LAN; bridges wireless and wired network segments.
10	Cloud	WAN Device	Represents the Internet or an external WAN; simulates remote connectivity between networks.

RESULT

Various networking components were successfully identified in Cisco Packet Tracer and their functions in a data communication system were studied.

AIM

To connect two PCs using a switch in Cisco Packet Tracer and verify communication between them using the ping command.

PROCEDURE**Step 1:** Open Cisco Packet Tracer

- Launch Cisco Packet Tracer from the desktop shortcut.
- Log in with your NetAcad credentials. A blank workspace appears.

Step 2: Add Devices to the Workspace

- In the Device Category Panel at the bottom left, click on End Devices.
- Drag and drop PC0 onto the left side of the workspace.
- Drag and drop PC1 onto the right side of the workspace.
- Click on Network Devices in the category panel.
- Select Switches from the sub-category. Drag and drop a Switch-2960 to the center of the workspace between the two PCs.

Step 3: Connect Devices Using Cables

- In the connections category (the lightning bolt icon at the bottom), select Copper Straight-Through cable.
- Click on PC0, select FastEthernet0 from the port list.
- Then click on the Switch, and select FastEthernet0/1 as the switch port.
- A green link light confirms a successful connection.
- Repeat: select Copper Straight-Through cable again. Click PC1 → FastEthernet0, then click Switch → FastEthernet0/2.
- Verify both connections show green lights at both ends.

Step 4: Assign IP Address to PC0

- Double-click on PC0 to open its configuration window.
- Go to the Desktop tab and click on IP Configuration.
- Select Static radio button.
- In the IP Address field, enter: 192.168.1.1
- In the Subnet Mask field, enter: 255.255.255.0
- Leave Default Gateway blank for now. Close the IP Configuration window.

Step 5: Assign IP Address to PC1

- Double-click on PC1. Go to Desktop → IP Configuration.
- Select Static. Enter IP Address: 192.168.1.2 and Subnet Mask: 255.255.255.0
- Close the window.

Step 6: Test Connectivity Using Ping

- Double-click on PC0. Go to Desktop → Command Prompt.
- The command prompt window opens. Type the following command and press Enter:

```
ping 192.168.1.2
```

- Observe the output. You should see four lines each beginning with 'Reply from 192.168.1.2' indicating successful communication.
- If you see 'Request timed out', re-check the IP addresses and cable connections.

Step 7: Verify from PC1 as well

- Double-click PC1 → Desktop → Command Prompt.
- Type:

```
ping 192.168.1.1
```

- Confirm replies are received from PC0. This verifies bidirectional communication.

Step 8: Record Ping Results

Ping Attempt	From	To	Result
1	PC0 (192.168.1.1)	PC1 (192.168.1.2)	Reply received
2	PC0 (192.168.1.1)	PC1 (192.168.1.2)	Reply received
3	PC1 (192.168.1.2)	PC0 (192.168.1.1)	Reply received
4	PC1 (192.168.1.2)	PC0 (192.168.1.1)	Reply received

RESULT

Two PCs were successfully connected using a Switch. Communication between them was verified using the ping command. All four replies were received successfully.

AIM

To study the working of ARP (Address Resolution Protocol) and observe packet flow between network devices using Simulation Mode in Cisco Packet Tracer.

PROCEDURE**Step 1:** Create the Network Topology

- Open Cisco Packet Tracer. Place the following devices on the workspace: PC0, PC1, Printer (from End Devices), and Switch-2960 (from Network Devices).
- Connect PC0 to the switch using Copper Straight-Through cable on FastEthernet0/1.
- Connect PC1 to the switch on FastEthernet0/2.
- Connect the Printer to the switch on FastEthernet0/3.
- Verify all link lights turn green.

Step 2: Assign IP Addresses to All Devices

- Double-click PC0 → Desktop → IP Configuration → Static.
- IP Address: 192.168.1.10 | Subnet Mask: 255.255.255.0. Close.
- Double-click PC1 → Desktop → IP Configuration → Static.
- IP Address: 192.168.1.20 | Subnet Mask: 255.255.255.0. Close.
- Double-click Printer → Desktop → IP Configuration → Static.
- IP Address: 192.168.1.30 | Subnet Mask: 255.255.255.0. Close.

Step 3: Clear the ARP Cache on PC0

- Double-click PC0 → Desktop → Command Prompt.
- Type the following command to view the current ARP table:

```
arp -a
```

- If any ARP entries exist, type the following to clear them:

```
arp -d
```

- Type arp -a again to confirm the table is now empty. Close command prompt.

Step 4: Switch to Simulation Mode

- At the bottom right of the Packet Tracer window, locate the Realtime / Simulation toggle.
- Click Simulation to switch to Simulation Mode. A Simulation Panel appears on the right side.
- In the Event List Filters section of the Simulation Panel, click 'Show All / None' to clear all filters, then tick only ARP and ICMP checkboxes so only these protocol packets are visible.

Step 5: Send a Ping from PC0 to the Printer

- Double-click PC0 → Desktop → Command Prompt.
- Type the ping command:

```
ping 192.168.1.30
```

- Do not press Enter yet. Return to the Simulation Panel.
- Now press Enter in the Command Prompt and immediately return to the workspace.

Step 6: Observe ARP Packet Flow Step by Step

- Click the 'Play' button or use 'Capture / Forward' (single-step button) in the Simulation Panel to advance one event at a time.
- Observe the following sequence of events in the workspace: ARP Request — PC0 broadcasts an ARP packet to all devices asking 'Who has 192.168.1.30? Tell 192.168.1.10'. The switch floods this to all ports. The Printer replies with an ARP Reply — 'I have 192.168.1.30. My MAC address is XX:XX:XX:XX:XX:XX'. PC0 stores this MAC in its ARP cache. ICMP Echo packets (ping) are then sent directly from PC0 to the Printer using the resolved MAC address.
- Click on any packet envelope in the workspace to see detailed packet information: Source IP, Destination IP, Source MAC, Destination MAC.

Step 7: Verify ARP Table After Communication

- Double-click PC0 → Desktop → Command Prompt.
- Type:

```
arp -a
```

- You should now see an entry mapping 192.168.1.30 to the Printer's MAC address — confirming ARP worked correctly.

OBSERVATION

Device	IP Address	MAC Resolved By ARP	ARP Entry in Table
PC0	192.168.1.10	—	Source
PC1	192.168.1.20	Yes (broadcast received)	—
Printer	192.168.1.30	Yes (replied to ARP)	Stored in PC0 cache

RESULT

The working of ARP protocol and packet flow between PC0 and the Printer was successfully observed using Simulation Mode. ARP request broadcast and unicast reply were captured and the MAC address was resolved and cached correctly.

AIM

To configure and test TCP-based protocols such as HTTP (web service) and Email (SMTP/POP3) using Cisco Packet Tracer.

PROCEDURE

PART A — Network Setup and IP Configuration

Step 1: Open Cisco Packet Tracer and Build the Topology

- Drag and place one Switch-2960, one Server (from End Devices → Servers), and three PCs (PC1, PC2, PC3) onto the workspace.
- Connect all devices to the switch using Copper Straight-Through cables.
- Connect: Server → Switch FastEthernet0/1, PC1 → Switch FastEthernet0/2, PC2 → Switch FastEthernet0/3, PC3 → Switch FastEthernet0/4.
- Wait for all port link lights to turn green (this may take a few seconds).

Step 2: Assign Static IP Address to the Server

- Double-click the Server → go to the Desktop tab → click IP Configuration.
- Select Static. Enter: IP Address: 192.168.11.5 | Subnet Mask: 255.255.255.0.
- Close the IP Configuration window.

Step 3: Assign Static IP Addresses to PC1, PC2, and PC3

- PC1 → Desktop → IP Configuration → Static: IP: 192.168.11.2 | Mask: 255.255.255.0
- PC2 → Desktop → IP Configuration → Static: IP: 192.168.11.3 | Mask: 255.255.255.0
- PC3 → Desktop → IP Configuration → Static: IP: 192.168.11.4 | Mask: 255.255.255.0

Step 4: Test Basic Connectivity

- Double-click PC1 → Desktop → Command Prompt. Type:

```
ping 192.168.11.5
```

- If 'Reply from 192.168.11.5' appears, the network is correctly configured. If 'Request timed out', re-check IP addresses and cable connections.

PART B — Configure and Test HTTP Service

Step 5: Enable HTTP Service on the Server

- Double-click the Server → go to the Services tab → click HTTP in the left panel.
- Ensure the HTTP service switch is turned ON. Also turn HTTPS ON.
- You will see a default web page listed (index.html). Click on index.html to edit it.

Step 6: Edit the Web Page Content

- In the text editor that opens, delete all existing HTML content.
- Type the following simple HTML:

```
<html>
  <body>
    <h1>Welcome to the TCP Web Server</h1>
```

```
<p>HTTP service is working correctly.</p>
</body>
</html>
```

- Click Save. The custom web page is now hosted on the server.

Step 7: Access the Web Page from PC2

- Double-click PC2 → Desktop tab → click Web Browser.
- In the address bar of the browser, type the Server's IP address:

```
192.168.11.5
```

- Press Enter or click Go. The web page should load, displaying the message: 'Welcome to the TCP Web Server'. This confirms HTTP is working over TCP.

PART C — Configure and Test Email Service (SMTP / POP3)

Step 8: Enable Email Service on the Server

- Double-click Server → Services → EMAIL (in the left panel).
- Turn SMTP service ON and POP3 service ON.
- In the Domain Name field, type: gmail.com and click Set.

Step 9: Create Email User Accounts

- In the User section of the EMAIL service, enter: Username: user1 | Password: 123 — then click + (Add button).
- Then enter: Username: user2 | Password: 123 — then click +.
- Verify both user1 and user2 appear in the user list.

Step 10: Configure Email Client on PC1

- Double-click PC1 → Desktop → Email (Mail icon).
- Fill in the configuration form: Your Name: User One | Email Address: user1@gmail.com | Incoming Mail Server (POP3): 192.168.11.5 | Outgoing Mail Server (SMTP): 192.168.11.5 | Username: user1 | Password: 123
- Click Save. The email client is now configured for user1.

Step 11: Configure Email Client on PC2

- Double-click PC2 → Desktop → Email.
- Fill in: Email Address: user2@gmail.com | Incoming Server: 192.168.11.5 | Outgoing Server: 192.168.11.5 | Username: user2 | Password: 123.
- Click Save.

Step 12: Send an Email from PC1 to PC2

- On PC1's email client, click Compose.
- Fill in the fields: To: user2@gmail.com | Subject: TCP Email Test | Message body: Hello, this is a test email using SMTP over TCP.
- Click Send.

Step 13: Receive Email on PC2

- On PC2's email client, click Receive.
- The inbox should update showing the email from user1@gmail.com with the subject 'TCP Email Test'. Click on the email to read the message.

RESULT

TCP-based services — HTTP (web page access) and Email (SMTP/POP3 send and receive) — were successfully configured and tested in Cisco Packet Tracer. Both services transmitted data reliably using TCP.

AIM

To configure and test the DNS (Domain Name System) service using UDP protocol in Cisco Packet Tracer.

PROCEDURE

Step 1: Build the Network Topology

- Open Cisco Packet Tracer. Drag and place: one Server, one PC, and one Switch-2960 onto the workspace.
- Connect Server → Switch FastEthernet0/1 using Copper Straight-Through cable.
- Connect PC → Switch FastEthernet0/2 using Copper Straight-Through cable.
- Verify both link lights are green.

Step 2: Assign IP Addresses

- Double-click Server → Desktop → IP Configuration → Static.
- IP Address: 192.168.11.5 | Subnet Mask: 255.255.255.0. Close.
- Double-click PC → Desktop → IP Configuration → Static.
- IP Address: 192.168.11.3 | Subnet Mask: 255.255.255.0. Close.

Step 3: Test Basic Connectivity

- Double-click PC → Desktop → Command Prompt. Type:

```
ping 192.168.11.5
```

- Confirm replies are received before proceeding to configure DNS.

Step 4: Enable and Configure DNS Service on the Server

- Double-click Server → Services tab → click DNS in the left panel.
- Toggle the DNS service switch to ON.
- In the Name field, type: www.google.com
- In the Address field, type: 192.168.11.5 (the server's own IP, simulating that google.com resolves to this server).
- Click Add. The entry 'www.google.com → 192.168.11.5' appears in the DNS records list.
- Add another entry: Name: www.myserver.com | Address: 192.168.11.5 → Click Add.

Step 5: Configure the PC to Use the DNS Server

- Double-click PC → Desktop → IP Configuration.
- In the DNS Server field, enter the server's IP address: 192.168.11.5
- Close the IP Configuration window.

Step 6: Test DNS Resolution Using nslookup

- Double-click PC → Desktop → Command Prompt. Type:

```
nslookup www.google.com
```

- Expected output: Server: 192.168.11.5 | Name: www.google.com | Address: 192.168.11.5
- This confirms the DNS server correctly resolved the domain name to an IP address using UDP port 53.

Step 7: Test DNS Using Web Browser

- Double-click PC → Desktop → Web Browser.
- In the address bar, type: `www.google.com` and press Enter.
- The PC sends a DNS query, receives the IP `192.168.11.5`, then makes an HTTP request to that IP. If HTTP is also enabled on the server, the web page will load — confirming end-to-end DNS and HTTP integration.

Step 8: Verify UDP Packet in Simulation Mode (Optional)

- Switch to Simulation Mode. Filter for DNS packets only.
- Repeat the `nslookup` command. Observe the DNS query packet traveling from PC to Server on UDP port 53, and the DNS reply returning with the resolved IP.

OBSERVATION

Query Type	Domain Name	DNS Server IP	Resolved IP	Status
nslookup	www.google.com	192.168.11.5	192.168.11.5	Success
Web Browser	www.google.com	192.168.11.5	192.168.11.5	Page Loaded
nslookup	www.myserver.com	192.168.11.5	192.168.11.5	Success

RESULT

DNS service was successfully configured on the server and tested from the PC using `nslookup` and web browser. The domain name was correctly resolved to an IP address using UDP protocol on port 53.

AIM

To configure OSPF (Open Shortest Path First) routing protocol between routers and verify connectivity between different networks using Cisco Packet Tracer.

PROCEDURE

Step 1: Open Cisco Packet Tracer and Create the Topology

- Drag and place two Routers (model 2911 or 1941) onto the workspace.
- Drag and place two PCs — PC1 on the left side and PC2 on the right side.
- Label the PCs PC1 (left) and PC2 (right) by clicking on their name labels.
- Label the routers Router1 (left) and Router2 (right) by clicking on their name labels.

Step 2: Connect All Devices with Cables

- Connect PC1 to Router1: use Copper Straight-Through cable. PC1 FastEthernet0 → Router1 GigabitEthernet0/0.
- Connect PC2 to Router2: use Copper Straight-Through cable. PC2 FastEthernet0 → Router2 GigabitEthernet0/0.
- Connect Router1 to Router2: use Copper Cross-Over cable (or select Automatic). Router1 GigabitEthernet0/1 → Router2 GigabitEthernet0/1.
- Verify link lights are green (router-to-router link will be amber initially — it turns green after interface configuration).

Step 3: Assign IP Address to PC1

- Double-click PC1 → Desktop → IP Configuration → Static.
- IP Address: 192.168.1.2 | Subnet Mask: 255.255.255.0 | Default Gateway: 192.168.1.1
- Close.

Step 4: Assign IP Address to PC2

- Double-click PC2 → Desktop → IP Configuration → Static.
- IP Address: 192.168.2.2 | Subnet Mask: 255.255.255.0 | Default Gateway: 192.168.2.1
- Close.

Step 5: Configure IP Addresses on Router1

- Double-click Router1 → CLI tab. Press Enter to start.
- Type the following commands exactly:

```
Router> enable
Router# configure terminal
Router(config)# hostname Router1
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# ip address 192.168.1.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface GigabitEthernet0/1
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

Step 6: Configure IP Addresses on Router2

- Double-click Router2 → CLI tab. Type:

```
Router> enable
Router# configure terminal
Router(config)# hostname Router2
Router2(config)# interface GigabitEthernet0/0
Router2(config-if)# ip address 192.168.2.1 255.255.255.0
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface GigabitEthernet0/1
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)# exit
```

Step 7: Configure OSPF on Router1

- Still in Router1's CLI. Type:

```
Router1(config)# router ospf 1
Router1(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router1(config-router)# exit
Router1(config)# exit
```

- Note: The wildcard mask 0.0.0.255 matches all hosts in the 192.168.1.0/24 network. The mask 0.0.0.3 matches the /30 link.

Step 8: Configure OSPF on Router2

- In Router2's CLI. Type:

```
Router2(config)# router ospf 1
Router2(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router2(config-router)# exit
Router2(config)# exit
```

Step 9: Verify OSPF Adjacency

- On either router CLI, type:

```
Router1# show ip ospf neighbor
```

- You should see Router2 listed as an OSPF neighbor with state 'FULL'. This means OSPF has established adjacency and exchanged routing information.

Step 10: Verify the Routing Table

- On Router1, type:

```
Router1# show ip route
```

- You should see an OSPF-learned route (marked with O) for the 192.168.2.0/24 network via 10.0.0.2.

Step 11: Test End-to-End Connectivity

- Double-click PC1 → Desktop → Command Prompt. Type:

```
ping 192.168.2.2
```

- You should receive replies from PC2 across the two different networks — confirming OSPF routing works correctly.

IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Purpose
PC1	FastEthernet0	192.168.1.2	255.255.255.0	End host on Network 1
Router1	GigabitEthernet0/0	192.168.1.1	255.255.255.0	Gateway for Network 1
Router1	GigabitEthernet0/1	10.0.0.1	255.255.255.252	WAN link to Router2
Router2	GigabitEthernet0/1	10.0.0.2	255.255.255.252	WAN link to Router1
Router2	GigabitEthernet0/0	192.168.2.1	255.255.255.0	Gateway for Network 2
PC2	FastEthernet0	192.168.2.2	255.255.255.0	End host on Network 2

RESULT

OSPF (Link State Routing) was configured successfully on both routers. Routing tables were updated dynamically using OSPF. End-to-end communication between PC1 (Network 1) and PC2 (Network 2) was verified successfully using the ping command.

AIM

To configure RIP (Routing Information Protocol — a Distance Vector routing protocol) on multiple routers and verify inter-network connectivity using Cisco Packet Tracer.

PROCEDURE

Step 1: Open Cisco Packet Tracer and Select Devices

- From the device panel, place the following devices on the workspace: 6 PCs (PC0 to PC5), 3 Switches (Switch-2960), 3 Routers (Router-2911).
- Arrange the routers in a horizontal line in the center. Place two PCs and one switch connected to each router to form three separate LAN segments.

Step 2: Connect Devices Using Cables

- LAN 1: Connect PC0 and PC1 to Switch0 using Copper Straight-Through cables. Connect Switch0 to Router0 FastEthernet0/0.
- LAN 2: Connect PC2 and PC3 to Switch1 using Copper Straight-Through cables. Connect Switch1 to Router1 FastEthernet0/0.
- LAN 3: Connect PC4 and PC5 to Switch2 using Copper Straight-Through cables. Connect Switch2 to Router2 FastEthernet0/0.
- WAN Links: Connect Router0 Serial2/0 to Router1 Serial2/0. Connect Router1 Serial3/0 to Router2 Serial2/0.
- Note: For Serial connections between routers, use Serial DCE/DTE cables. One router acts as DCE (clock source) and must have a clock rate configured.

Step 3: Assign IP Addresses to PCs

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1
PC4	192.168.30.2	255.255.255.0	192.168.30.1
PC5	192.168.30.3	255.255.255.0	192.168.30.1

- For each PC: Double-click → Desktop → IP Configuration → Static → enter IP, Mask, and Gateway as shown.

Step 4: Configure Router0

- Double-click Router0 → CLI tab. Type:

```
Router> enable
Router# configure terminal
Router(config)# hostname Router0
Router0(config)# interface FastEthernet0/0
Router0(config-if)# ip address 192.168.10.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface Serial2/0
```

```
Router0(config-if)# ip address 10.0.0.1 255.0.0.0
Router0(config-if)# clock rate 64000
Router0(config-if)# no shutdown
Router0(config-if)# exit
```

Step 5: Configure Router1

```
Router> enable
Router# configure terminal
Router(config)# hostname Router1
Router1(config)# interface FastEthernet0/0
Router1(config-if)# ip address 192.168.20.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface Serial2/0
Router1(config-if)# ip address 10.0.0.2 255.0.0.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface Serial3/0
Router1(config-if)# ip address 11.0.0.1 255.0.0.0
Router1(config-if)# clock rate 64000
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

Step 6: Configure Router2

```
Router> enable
Router# configure terminal
Router(config)# hostname Router2
Router2(config)# interface FastEthernet0/0
Router2(config-if)# ip address 192.168.30.1 255.255.255.0
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface Serial2/0
Router2(config-if)# ip address 11.0.0.2 255.0.0.0
Router2(config-if)# no shutdown
Router2(config-if)# exit
```

Step 7: Configure RIP on All Three Routers

- On Router0 CLI:

```
Router0(config)# router rip
Router0(config-router)# network 192.168.10.0
Router0(config-router)# network 10.0.0.0
Router0(config-router)# exit
```

- On Router1 CLI:

```
Router1(config)# router rip
Router1(config-router)# network 192.168.20.0
Router1(config-router)# network 10.0.0.0
Router1(config-router)# network 11.0.0.0
Router1(config-router)# exit
```

- On Router2 CLI:

```
Router2(config)# router rip
Router2(config-router)# network 192.168.30.0
```

```
Router2(config-router)# network 11.0.0.0
Router2(config-router)# exit
```

Step 8: Verify Routing Tables

- On any router CLI, type:

```
show ip route
```

- You should see RIP-learned routes marked with R in the routing table. Router0 should show routes to 192.168.20.0 and 192.168.30.0 learned via RIP.

Step 9: Test Inter-Network Connectivity

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.20.2
```

- Then also test:

```
ping 192.168.30.2
```

- Both pings should receive replies, confirming that RIP has enabled inter-network routing successfully.

RESULT

RIP (Distance Vector Routing Protocol) was configured successfully on all three routers. Routing tables were populated dynamically by RIP. End-to-end communication across all three networks was verified using the ping command.

AIM

To install and configure Wireshark network protocol analyzer on a Windows operating system in order to capture and analyze network packets.

PROCEDURE

Step 1: Open the Official Wireshark Website

- Open any web browser (Chrome, Firefox, or Edge).
- Navigate to: <https://www.wireshark.org>
- The official Wireshark homepage loads. Do NOT download from third-party websites.

Step 2: Download the Windows Installer

- Click on the Download button prominently displayed on the homepage.
- The download page shows the latest stable release. Under Windows, click 'Windows x64 Installer' for 64-bit systems (recommended) or 'Windows x32 Installer' for 32-bit.
- The file (e.g., Wireshark-4.x.x-x64.exe) downloads to your Downloads folder.
- Verify the file size is approximately 70–80 MB before proceeding.

Step 3: Run the Installer as Administrator

- Open File Explorer and navigate to the Downloads folder.
- Right-click on the downloaded .exe file and select 'Run as administrator' for elevated permissions.
- If a User Account Control (UAC) prompt appears, click Yes to allow the installation.

Step 4: Proceed Through the Setup Wizard

- The Wireshark Setup Wizard opens. Click Next to begin.
- The License Agreement (GNU General Public License) appears. Read it and click 'I Agree' to accept and continue.

Step 5: Select Components to Install

- The component selection screen appears. Ensure the following are checked: Wireshark (main application), TShark (command-line version), Plugins & Extensions, User's Guide (optional).
- Leave the default selections and click Next.

Step 6: Choose Additional Tasks and File Associations

- On the 'Additional Tasks' screen, check 'Associate .pcap files with Wireshark' so captured files open automatically.
- Optionally check 'Create Shortcut on Desktop'. Click Next.

Step 7: Select Installation Folder

- The default installation path is C:\Program Files\Wireshark. Leave this as default.
- Click Next.

Step 8: Install Npcap (Packet Capture Driver)

- A prompt asks whether to install Npcap — the packet capture library that allows Wireshark to capture live network traffic.

- Click Install Npcap (strongly recommended). The Npcap installer launches separately.
- In the Npcap setup: Accept the License Agreement → click I Agree.
- On the installation options, leave defaults (WinPcap API-compatible mode may be checked). Click Install.
- Npcap installs. Click Finish when complete. You return to the Wireshark installer.

Step 9: Complete Wireshark Installation

- Click Install in the Wireshark wizard. The progress bar shows installation progress.
- Wait for all components to install (approximately 1–3 minutes).
- Click Next, then Finish to complete the setup. Leave 'Run Wireshark' checked.

Step 10: Launch and Verify Wireshark

- Wireshark opens. The Welcome screen shows a list of available network interfaces (e.g., Wi-Fi, Ethernet).
- Verify that network interfaces are listed — this confirms Npcap is working correctly.
- Double-click on your active interface (e.g., Wi-Fi) to start a test capture.
- Packets should appear in the packet list pane, confirming Wireshark is fully functional.
- Click the red Stop button to stop the capture.

OBSERVATION

Step	Observation
1	Installer downloaded successfully from wireshark.org
2	Setup wizard launched without errors
3	Npcap installed successfully as packet capture driver
4	Wireshark installed with all selected components
5	Network interfaces listed on Wireshark welcome screen
6	Live packet capture started and packets visible in real time

RESULT

Wireshark was successfully installed on the Windows operating system. Npcap driver was configured, and the application is ready to capture and analyze live network traffic.

AIM

To install and configure Wireshark network protocol analyzer on a Linux operating system to capture and analyze network packets.

PROCEDURE

Step 1: Open the Terminal

- On the Linux desktop, press Ctrl + Alt + T to open the Terminal, or right-click the desktop and select 'Open Terminal'.
- The terminal prompt appears (e.g., username@hostname:~\$).

Step 2: Update the Package List

- Before installing any software, update the package index to get the latest versions:

```
sudo apt update
```

- Enter your user password when prompted. Wait for the update to complete.

Step 3: Upgrade Existing Packages (Optional but Recommended)

- Type:

```
sudo apt upgrade -y
```

- This ensures all existing packages are up to date before installing Wireshark.

Step 4: Install Wireshark

- Install Wireshark using the apt package manager:

```
sudo apt install wireshark -y
```

- The system downloads and installs Wireshark and all its dependencies.

Step 5: Configure Non-Root Packet Capture Permission

- During installation, a dialog box appears: 'Should non-superusers be able to capture packets?'
- Use the arrow keys to select Yes and press Enter. This allows your normal user account to capture packets without running Wireshark as root every time.

Step 6: Add Your User to the Wireshark Group

- Even after selecting Yes above, you must manually add your user to the wireshark group. Type the following command, replacing 'your_username' with your actual Linux username:

```
sudo usermod -aG wireshark $USER
```

- This adds your user account to the wireshark group which has capture permissions.

Step 7: Apply Group Changes

- Log out and log back in, or restart the system for the group change to take effect:

```
sudo reboot
```

- Alternatively, apply the change immediately without rebooting:

```
newgrp wireshark
```

Step 8: Launch Wireshark

- Launch Wireshark from the terminal:

```
wireshark &
```

- The & symbol runs it in the background so the terminal remains available. Alternatively, search for Wireshark in the application menu and click its icon.

Step 9: Verify Network Interfaces Are Listed

- The Wireshark Welcome screen opens. It should display all available network interfaces (e.g., eth0, wlan0, lo).
- If interfaces appear, packet capture permissions are correctly configured.
- Double-click any active interface to start a live capture. Packets should appear in real time.
- Click the red Stop button to halt the capture.

Step 10: Verify Wireshark Version

- In the terminal, verify the installed version:

```
wireshark --version
```

- The version number and build details are displayed, confirming successful installation.

OBSERVATION

Step	Observation / Command Result
sudo apt update	Package list updated successfully
sudo apt install wireshark	Wireshark installed with all dependencies
Non-root capture dialog	Selected 'Yes' — non-superusers allowed
usermod command	User added to wireshark group
wireshark command	Wireshark GUI launched successfully
Network interfaces	eth0 / wlan0 listed; live capture started

RESULT

Wireshark was successfully installed on the Linux operating system using apt package manager. The user was added to the wireshark group for non-root packet capture. The application is ready to capture and analyze network traffic.

AIM

To capture and analyze HTTP and TCP packets using the network protocol analyzer Wireshark.

PROCEDURE

Step 1: Launch Wireshark

- Open Wireshark from the desktop or Start Menu. The Welcome screen shows available network interfaces.

Step 2: Select the Active Network Interface

- Identify your active interface: if connected via Wi-Fi, select Wi-Fi (wlan0). If connected via Ethernet cable, select Ethernet (eth0).
- Look for the interface with a moving traffic waveform graph — that indicates active traffic.

Step 3: Start Packet Capture

- Double-click on the active interface, or click the interface once and then click the blue shark fin icon (Start Capturing Packets) in the toolbar.
- Packet capture begins. The packet list pane fills with packets in real time.

Step 4: Generate HTTP Traffic

- Open a web browser (Chrome, Firefox, etc.).
- Navigate to a website that uses HTTP (not HTTPS). A good option for testing is:

```
http://httpforever.com
```

- Or use <http://neverssl.com> — these are plain HTTP sites without HTTPS redirection.
- The browser sends HTTP GET requests and receives HTTP responses.

Step 5: Apply HTTP Display Filter in Wireshark

- Return to Wireshark. In the Display Filter bar at the top, type:

```
http
```

- Press Enter. Wireshark now shows only HTTP packets.
- Observe HTTP GET requests (from your PC to the server) and HTTP 200 OK responses (from the server back to your PC).
- Click on any HTTP GET packet to expand it in the Packet Details pane below. Look at the Hypertext Transfer Protocol section to see the request method, URL, and headers.

Step 6: Apply TCP Display Filter

- Clear the HTTP filter by clicking the X button beside the filter bar.
- Type in the filter bar:

```
Tcp
```

- Press Enter. All TCP packets are now displayed.
- Look for the three-way handshake packets: SYN (client initiates), SYN-ACK (server acknowledges), ACK (client confirms).
- Click on any TCP packet to see its details: Source Port, Destination Port, Sequence Number, Acknowledgment Number, Flags.

Step 7: Filter HTTP Traffic to a Specific Server

- For more focused analysis, combine filters:

```
http and ip.addr == <server IP address>
```

- Replace <server IP address> with the website's IP address to see only traffic to that specific server.

Step 8: Stop the Capture and Save

- Click the red square Stop button in the toolbar to halt packet capture.
- Go to File → Save As. Enter a filename (e.g., http_tcp_capture) and save as .pcap or .pcapng format.

OBSERVATION

No.	Protocol	Source IP	Destination IP	Info / Description
1	—	—	—	—
2	—	—	—	—
3	—	—	—	—
4	—	—	—	—
5	—	—	—	—

RESULT

HTTP and TCP packets were successfully captured and analyzed using Wireshark. HTTP GET requests and responses were identified, and the TCP three-way handshake (SYN, SYN-ACK, ACK) was observed in the packet details.

AIM

To capture and analyze DNS and ARP packets using the network protocol analyzer Wireshark.

PROCEDURE

Step 1: Open Wireshark and Select Network Interface

- Launch Wireshark. On the Welcome screen, identify the active interface (Wi-Fi or Ethernet — look for the moving traffic graph).
- Double-click the active interface to begin capture immediately.

Step 2: Clear Existing DNS Cache Before Capture

- Before generating DNS traffic, clear the DNS cache so fresh DNS queries are generated. Open Command Prompt (Windows) or Terminal (Linux).
- On Windows:

```
ipconfig /flushdns
```

- On Linux:

```
sudo systemd-resolve --flush-caches
```

Step 3: Generate DNS Traffic

- Open a web browser and type a website address, for example:

```
www.google.com
```

- Press Enter. Before the browser loads the page, the operating system sends a DNS query to the configured DNS server to resolve www.google.com to an IP address.

Step 4: Filter and Observe DNS Packets in Wireshark

- Return to Wireshark. In the Display Filter bar, type:

```
Dns
```

- Press Enter. Only DNS packets are shown.
- Look for: DNS Query packet — from your PC to the DNS server. It contains the domain name being queried (e.g., www.google.com). DNS Response packet — from the DNS server to your PC. It contains the resolved IP address (e.g., 142.250.x.x for Google).
- Click on the DNS query packet. In the Packet Details pane, expand 'Domain Name System (query)' to see the queried domain name.
- Click on the DNS response packet. Expand the 'Answers' section to see the resolved IP address.

Step 5: Generate ARP Traffic

- Clear the Wireshark filter bar (click X). Open Command Prompt or Terminal.
- Ping a device on your local network to generate ARP traffic. For example:

```
ping 192.168.1.1
```

- Before the first ping can be sent, your PC must resolve the IP 192.168.1.1 to a MAC address using ARP.

Step 6: Filter and Observe ARP Packets

- In Wireshark's Display Filter bar, type:

Arp

- Press Enter. ARP packets are displayed.
- Identify: ARP Request — 'Who has 192.168.1.1? Tell 192.168.x.x' — broadcast to all devices. ARP Reply — '192.168.1.1 is at AA:BB:CC:DD:EE:FF' — unicast reply with the MAC address.
- Click on the ARP Request packet. In the Packet Details pane, expand 'Address Resolution Protocol': Sender MAC, Sender IP, Target MAC (all zeros in request), Target IP (the IP being queried).
- Click on the ARP Reply packet and expand to see the resolved MAC address.

Step 7: Stop the Capture

- Click the red Stop button. Optionally save the capture as File → Save As → dns_arp_capture.pcap.

OBSERVATION

Packet Type	Protocol	Source	Destination	Key Information
DNS Query	UDP Port 53	Your PC IP	DNS Server IP	Queried domain: www.google.com
DNS Response	UDP Port 53	DNS Server IP	Your PC IP	Resolved IP address returned
ARP Request	ARP	Your MAC / IP	Broadcast (FF:FF:FF:FF:FF:FF)	Who has 192.168.1.1?
ARP Reply	ARP	Target MAC / IP	Your MAC / IP	MAC address of target device

RESULT

DNS and ARP packets were successfully captured and analyzed using Wireshark. The DNS query and response showing domain-to-IP resolution, and the ARP request and reply showing IP-to-MAC resolution, were clearly observed.

AIM

To use Wireshark for capturing and analyzing unencrypted HTTP traffic from a vulnerable website, and to demonstrate how sensitive information such as usernames and passwords can be intercepted through eavesdropping.

PROCEDURE

Step 1: Launch Wireshark

- Open Wireshark from the desktop or application menu. The Welcome screen with network interfaces appears.

Step 2: Select the Active Network Interface

- Identify the active interface (the one showing a moving waveform). Click once to select it.

Step 3: Apply a Capture Filter for HTTP

- Before starting capture, apply a capture filter to reduce noise. In the 'Capture Filter' text box on the main screen, type:

```
port 80
```

- This captures only traffic on port 80 (HTTP), filtering out HTTPS and other traffic.

Step 4: Start Packet Capture

- Click the blue shark fin Start button in the toolbar to begin capturing.

Step 5: Navigate to a Vulnerable HTTP Website

- Open a web browser and navigate to a deliberately vulnerable HTTP-based website for testing purposes. Recommended safe practice sites:
- Acunetix Test Site: <http://testphp.vulnweb.com> — a legally accessible test site for security training.
- DVWA (Damn Vulnerable Web Application) — if set up locally in a lab environment.
- On the test website, find a login form and enter sample test credentials: Username: admin | Password: test123.
- Click the Login or Submit button on the form.

Step 6: Stop Packet Capture

- Return to Wireshark and click the red Stop button to halt packet capture.

Step 7: Filter for HTTP POST Requests

- In the Display Filter bar, type:

```
http.request.method == POST
```

- Press Enter. This shows only HTTP POST requests — the type of request generated when a form is submitted with login credentials.

Step 8: Locate the Login Request Packet

- In the filtered packet list, identify the POST request sent to the login page (usually identifiable by its destination URI, e.g., /login.php or /userinfo.php).
- Click on that packet to select it.

Step 9: Analyze the HTTP POST Request Details

- In the Packet Details pane, expand the section labeled 'Hypertext Transfer Protocol'.
- Look for the 'HTML Form URL Encoded' section or the 'Line-based text data' section.
- Expand it to find form field values. You should be able to clearly read: `uname=admin` (or whatever username was entered) and `pass=test123` (or whatever password was entered) — in completely plain text.

Step 10: Use 'Follow TCP Stream' for Full Context

- Right-click on the same POST packet. Select 'Follow' → 'TCP Stream'.
- A new window opens showing the complete HTTP conversation between the browser and the server in readable text. The username and password fields are visible in plain text within the HTTP request body.

Step 11: Save the Capture for Documentation

- Close the TCP Stream window. Go to File → Save As.
- Enter filename: `http_credential_capture.pcap` and click Save.

OBSERVATION

Observation Point	Finding
Protocol used	HTTP (unencrypted, port 80)
Request method	POST — form submission with login data
Username field	Visible in plaintext in the packet body
Password field	Visible in plaintext in the packet body
Encryption	None — all data transmitted in cleartext
Security risk	Credentials can be intercepted by any device on the same network

RESULT

Unencrypted HTTP traffic was successfully captured and analyzed. The username and password submitted through the HTTP login form were visible in plaintext within the captured packets, demonstrating the critical importance of using HTTPS (encrypted) connections for all sensitive data transmission.

AIM

To capture and analyze the TCP three-way handshake process using the network protocol analyzer Wireshark.

PROCEDURE**Step 1: Launch Wireshark and Start Capture**

- Open Wireshark. Select the active network interface. Click the Start Capturing button (blue shark fin).

Step 2: Generate a Fresh TCP Connection

- Open a web browser. Navigate to a new website you have not visited recently (so a fresh TCP connection is established):

```
http://httpforever.com
```

- A fresh TCP connection is created between your browser and the web server — this includes the three-way handshake.

Step 3: Stop the Capture and Apply TCP Filter

- Return to Wireshark. Click the red Stop button.
- In the Display Filter bar, type:

```
Tcp
```

- Press Enter. Only TCP packets are now displayed.

Step 4: Locate the Three-Way Handshake Sequence

- Scroll to the top of the packet list. Look for the following three consecutive packets in order: Packet 1 — SYN: Source = your PC, Destination = web server. In the Info column: 'SYN'. Packet 2 — SYN-ACK: Source = web server, Destination = your PC. In the Info column: 'SYN, ACK'. Packet 3 — ACK: Source = your PC, Destination = web server. In the Info column: 'ACK'.

Step 5: Analyze the SYN Packet (Packet 1)

- Click on the SYN packet to select it.
- In the Packet Details pane, expand 'Transmission Control Protocol'.
- Observe: Source Port (random high port, e.g., 54321), Destination Port (80 for HTTP), Sequence Number (random initial sequence number), Flags: SYN = 1 (only SYN flag is set).

Step 6: Analyze the SYN-ACK Packet (Packet 2)

- Click on the SYN-ACK packet.
- In the Packet Details pane, expand 'Transmission Control Protocol'.
- Observe: Source Port (80), Destination Port (your random port), Sequence Number (server's ISN), Acknowledgment Number (client ISN + 1), Flags: SYN = 1, ACK = 1.

Step 7: Analyze the ACK Packet (Packet 3)

- Click on the ACK packet.
- Expand 'Transmission Control Protocol'.

- Observe: Source Port (your random port), Destination Port (80), Sequence Number (client ISN + 1), Acknowledgment Number (server ISN + 1), Flags: ACK = 1 (only ACK flag set). The connection is now established.

Step 8: Use 'Follow TCP Stream' to See the Full Session

- Right-click on any of the three handshake packets. Select 'Follow' → 'TCP Stream'.
- The complete TCP session is shown — from the handshake through the HTTP GET request and server response. This gives a complete picture of the TCP connection lifecycle.

Step 9: Filter Specifically for Handshake Packets

- For a cleaner view showing only SYN and SYN-ACK packets, use the filter:

```
tcp.flags.syn == 1
```

- This shows all SYN and SYN-ACK packets, helping you identify all connection establishment events in the capture.

OBSERVATION — TCP Three-Way Handshake

Step	Packet Type	From	To	TCP Flags	Purpose
1	SYN	Client	Server	SYN=1, ACK=0	Client requests connection
2	SYN-ACK	Server	Client	SYN=1, ACK=1	Server accepts and acknowledges
3	ACK	Client	Server	SYN=0, ACK=1	Client confirms — connection established

RESULT

The TCP three-way handshake (SYN → SYN-ACK → ACK) was successfully captured and analyzed using Wireshark. Each packet's TCP flags, sequence numbers, and acknowledgment numbers were examined, confirming the connection establishment process between client and server.

AIM

To analyze network traffic statistics using the network protocol analyzer Wireshark.

PROCEDURE

Step 1: Launch Wireshark and Select Interface

- Open Wireshark. On the Welcome screen, double-click the active network interface to begin capture.

Step 2: Generate Varied Network Traffic

- To ensure diverse protocol statistics, generate different types of traffic: Open a browser and visit two or three websites (generates HTTP/HTTPS, DNS, TCP). Open a command prompt and run:

```
ping 8.8.8.8 -n 10
```

- This generates ICMP packets. Also ping your gateway:

```
ping 192.168.1.1 -n 5
```

- Wait approximately 30 seconds while browsing to accumulate meaningful traffic data.

Step 3: Stop the Capture

- Click the red Stop button in the Wireshark toolbar to stop packet capture.

Step 4: View Protocol Hierarchy Statistics

- In the Wireshark menu bar, click Statistics → Protocol Hierarchy.
- A window opens showing a tree-view of all protocols detected in the capture, with columns: Protocol, Packets (count), Bytes, and percentage of total traffic.
- Observe the hierarchy: at the top is the Frame / Ethernet layer. Below it is IPv4 or IPv6. Below IP you see TCP, UDP, ICMP. Below TCP you see HTTP, TLS (HTTPS), etc.
- Record the number of packets and percentage for each major protocol in the observation table. Close this window.

Step 5: View Conversations Statistics

- Go to Statistics → Conversations.
- The Conversations window opens with multiple tabs: Ethernet (shows MAC-to-MAC conversations), IPv4 (shows IP-to-IP conversations), TCP (shows port-to-port conversations), UDP.
- Click the IPv4 tab. Observe the list of communication pairs (your PC's IP and the remote servers it connected to), along with packet counts and byte totals for each conversation.
- Click the TCP tab to see specific port-level conversations. Close this window.

Step 6: View Endpoints Statistics

- Go to Statistics → Endpoints.
- The Endpoints window shows all network endpoints (individual IP addresses and MAC addresses) that appeared in the capture.
- Click the IPv4 tab. Look for your PC's IP address and remote server addresses. Each entry shows how many packets were sent/received and total bytes.
- Close the Endpoints window.

Step 7: View I/O Graphs

- Go to Statistics → I/O Graph.
- A graph appears showing network throughput over time. The Y-axis shows packets per interval, the X-axis shows time.
- Observe peaks in traffic that correspond to when you browsed websites or ran pings. This gives a visual picture of network activity over the capture duration. Close the graph.

Step 8: View Capture File Properties

- Go to Statistics → Capture File Properties.
- This shows: Total packets captured, Capture duration, Average packets per second, Average bytes per second.

Step 9: Record Observations

S.No	Protocol	Packets Count	% of Total	Bytes	
1	Ethernet	—	—	—	—
2	IPv4	—	—	—	—
3	TCP	—	—	—	—
4	UDP	—	—	—	—
5	ICMP	—	—	—	—
6	HTTP	—	—	—	—
7	DNS	—	—	—	—

RESULT

Network traffic statistics were successfully analyzed using Wireshark. Protocol hierarchy, conversation details, and endpoint information were examined using the Statistics menu. The distribution of protocols across the captured network traffic was observed and recorded.

AIM

To capture network packets and save the captured data as a file for later analysis using Wireshark.

PROCEDURE**Step 1: Launch Wireshark and Select Interface**

- Open Wireshark. On the Welcome screen, identify your active network interface (the one with the moving waveform graph).
- Double-click the interface or click it once and press the Start button to begin capture.

Step 2: Generate Network Traffic

- While Wireshark is capturing, generate some network activity.
- Open a web browser and visit any website (e.g., www.google.com).
- Open Command Prompt / Terminal and run:

```
ping 8.8.8.8 -n 5
```

- Observe that packets are appearing in the Wireshark packet list in real time.

Step 3: Stop the Capture

- After 15–30 seconds of capturing, click the red square Stop button in the Wireshark toolbar.
- The packet list is now static — no new packets are added.

Step 4: Save the Capture File

- In the Wireshark menu bar, go to File → Save As.
- The Save dialog opens. Navigate to your Desktop or a preferred folder.
- In the 'File name' field, type a meaningful name, for example: my_first_capture
- In the 'Save as type' dropdown, select 'Wireshark/tcpdump/... pcap' (.pcap) or 'Wireshark/... pcapng' (.pcapng). The .pcapng format is recommended as it stores more metadata.
- Click Save. The file is saved to the selected location.

Step 5: Verify the Saved File

- Open File Explorer (Windows) or the file manager (Linux). Navigate to where you saved the file.
- Confirm the file exists with a .pcap or .pcapng extension and a non-zero file size (typically several KB to MB depending on captured traffic).

Step 6: Close and Reopen Wireshark

- Close Wireshark completely (File → Quit or press the X button).
- Reopen Wireshark from the Start Menu or desktop shortcut.

Step 7: Open the Saved Capture File

- In the Wireshark menu bar, go to File → Open.
- Browse to the location where you saved the .pcap file.
- Select the file and click Open.
- Alternatively, you can drag and drop the .pcap file directly onto the Wireshark window to open it.

Step 8: Verify Loaded Packets

- The saved packets are now loaded in the packet list pane. You should see the same packets as before.
- Apply a filter to verify specific packet types, for example:

Icmp

- The ICMP ping packets you generated should appear. This confirms the file was saved and loaded correctly.

Step 9: Record File Details

Action	Details	Observation
Packet Capture	Interface selected and capture started	Packets visible in real time
Capture Stopped	Stop button clicked	Packet count frozen
File Saved	File → Save As → .pcapng	File created on disk
File Reopened	File → Open → selected file	All packets reloaded for analysis

RESULT

Network packets were successfully captured, saved as a .pcap/.pcapng file, and reopened in Wireshark for analysis. The saved file retained all captured packet details.

AIM

To capture network packets and identify the source and destination IP addresses of each packet using Wireshark.

PROCEDURE**Step 1: Launch Wireshark**

- Open Wireshark. Select the active network interface from the Welcome screen (look for the waveform animation).
- Double-click the interface to start packet capture immediately.

Step 2: Generate Specific Network Traffic

- Open Command Prompt (Windows) or Terminal (Linux).
- Ping a known IP address to generate traceable ICMP traffic:

```
ping 8.8.8.8 -n 10
```

- Also open a web browser and visit www.google.com to generate DNS and TCP/HTTP traffic.

Step 3: Stop the Capture

- After 20 seconds, click the red Stop button in Wireshark.

Step 4: Locate the Source and Destination IP Columns

- In the Packet List Pane (the top panel), locate the column headers: No. (packet number), Time, Source, Destination, Protocol, Length, Info.
- The Source column shows the IP address of the device that sent the packet.
- The Destination column shows the IP address of the device that should receive the packet.

Step 5: Apply an ICMP Filter to Isolate Ping Traffic

- In the Display Filter bar, type:

```
Icmp
```

- Press Enter. Only ICMP (ping) packets are shown. These are easy to analyze because the Source and Destination IPs alternate: packets going to 8.8.8.8 show your IP as Source and 8.8.8.8 as Destination. Reply packets show 8.8.8.8 as Source and your IP as Destination.

Step 6: Click a Packet and View Detailed IP Information

- Click on any ICMP packet. In the Packet Details pane (middle panel), expand 'Internet Protocol Version 4'.
- Observe the fields: Version (4 for IPv4), Header Length, Time to Live (TTL), Protocol (1 for ICMP), Source Address (the sender's IP), Destination Address (the receiver's IP).

Step 7: Use DNS Filter to See Name-to-IP Mapping

- Clear the ICMP filter and type:

```
Dns
```

- DNS packets show your PC's IP as Source (for queries) and the DNS server's IP as Source (for responses). The Info column shows the queried domain name and the resolved IP in the response.

Step 8: Sort by Source or Destination Column

- Click on the 'Source' column header to sort packets by source IP. This groups all packets from the same source together — useful for identifying which devices are most active.
- Click on 'Destination' to sort by destination IP.

Step 9: Record Source and Destination IPs

Packet No.	Protocol	Source IP Address	Destination IP Address
1	ICMP	Your PC IP	8.8.8.8
2	ICMP	8.8.8.8	Your PC IP
3	DNS	Your PC IP	DNS Server IP
4	DNS	DNS Server IP	Your PC IP
5	TCP	Your PC IP	Web Server IP
6	TCP	Web Server IP	Your PC IP

RESULT

Source and destination IP addresses were successfully identified from captured packets using Wireshark. Packet-level IP addressing was observed for ICMP, DNS, and TCP protocols.

AIM

To create a Bus Topology using Cisco Packet Tracer and verify connectivity between the computers using the Ping command.

PROCEDURE

Step 1: Open Cisco Packet Tracer

- Launch Cisco Packet Tracer. Log in with NetAcad credentials. A blank workspace opens.

Step 2: Add Devices to the Workspace

- From End Devices, drag and place four PCs: PC0, PC1, PC2, and PC3 onto the workspace.
- From Network Devices, drag and place Hub-PT onto the workspace.
- Arrange the PCs in a horizontal row and place the Hub-PT appropriately.

Step 3: Connect Devices Using Copper Straight-Through Cable

- Select Copper Straight-Through cable from the Connections category (lightning bolt icon).
- Connect each PC to the central device / adjacent PC as described in the topology.
- Verify that all link lights turn green. If a link shows red, check that the correct ports are selected.

Step 4: Assign IP Addresses to All PCs

- For each PC: Double-click the PC → Desktop tab → IP Configuration → select Static.
- Enter the IP address and subnet mask from the table below:

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Step 5: Understand Bus Topology Physical Layout

- In a Bus Topology, all devices connect to a single shared communication line. In Cisco Packet Tracer, a Hub simulates this because it broadcasts all received traffic to every connected port — exactly like a shared bus.
- Confirm your topology: PC0 → Hub Port0, PC1 → Hub Port1, PC2 → Hub Port2, PC3 → Hub Port3.

Step 6: Test Connectivity From PC0 to All Other PCs

- Double-click PC0 → Desktop → Command Prompt.
- Type each ping command and press Enter after each:

```
ping 192.168.1.2
```

```
ping 192.168.1.3
```

```
ping 192.168.1.4
```

- For each ping, you should receive: 'Reply from 192.168.x.x: bytes=32 time<1ms TTL=128'

- If 'Request timed out' appears: re-check the IP configuration and cable connections.

Step 7: Observe Broadcast Behavior Using Simulation Mode

- Switch to Simulation Mode (bottom right). Filter for ICMP only.
- Send a ping from PC0. Observe that the Hub broadcasts the ARP request to ALL connected PCs — this is the defining characteristic of Bus topology. Every device receives every transmission.

Step 8: Record Observation

Ping Attempt	From PC	To PC	Result
1	PC0	PC1	Reply received / Timed out
2	PC0	PC2	Reply received / Timed out
3	PC0	PC3	Reply received / Timed out
4	PC1	PC2	Reply received / Timed out

RESULT

Bus Topology was successfully created in Cisco Packet Tracer using a Hub as the shared communication medium. Connectivity between all four PCs was verified using the ping command. All reply packets were received successfully.

AIM

To create a Star Topology using Cisco Packet Tracer and verify connectivity between the computers using the Ping command.

PROCEDURE**Step 1:** Open Cisco Packet Tracer

- Launch Cisco Packet Tracer. Log in with NetAcad credentials. A blank workspace opens.

Step 2: Add Devices to the Workspace

- From End Devices, drag and place four PCs: PC0, PC1, PC2, and PC3 onto the workspace.
- From Network Devices, drag and place Switch-2960 onto the workspace.
- Arrange the PCs in positions around a central switch and place the Switch-2960 appropriately.

Step 3: Connect Devices Using Copper Straight-Through Cable

- Select Copper Straight-Through cable from the Connections category (lightning bolt icon).
- Connect each PC to the central device / adjacent PC as described in the topology.
- Verify that all link lights turn green. If a link shows red, check that the correct ports are selected.

Step 4: Assign IP Addresses to All PCs

- For each PC: Double-click the PC → Desktop tab → IP Configuration → select Static.
- Enter the IP address and subnet mask from the table below:

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Step 5: Understand Star Topology Physical Layout

- In a Star Topology, all devices connect to a central switch. The switch uses a MAC address table to intelligently forward frames only to the intended destination port — unlike a hub which broadcasts to all.
- Confirm connections: PC0 → Switch FastEthernet0/1, PC1 → Switch FastEthernet0/2, PC2 → Switch FastEthernet0/3, PC3 → Switch FastEthernet0/4.
- All four link lights on the switch should be green.

Step 6: Verify the Switch MAC Address Table (Optional)

- Double-click the Switch → CLI tab.
- After the PCs have communicated, type:

```
Switch# show mac address-table
```

- The MAC table shows which MAC address is connected to which port. This demonstrates that the switch learns device locations dynamically.

Step 7: Test Connectivity From PC0

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.1.2  
ping 192.168.1.3  
ping 192.168.1.4
```

- All pings should receive 4 replies each, confirming successful connectivity through the switch.

Step 8: Compare with Bus Topology

- In Simulation Mode, observe that the switch sends ARP replies only to the requesting PC — not to all PCs. This intelligent unicast forwarding is the advantage of Star Topology with a switch over Bus Topology with a hub.

Step 9: Record Observation

Ping Attempt	From PC	To PC	Result
1	PC0	PC1	Reply received / Timed out
2	PC0	PC2	Reply received / Timed out
3	PC0	PC3	Reply received / Timed out
4	PC1	PC2	Reply received / Timed out

RESULT

Star Topology was successfully created in Cisco Packet Tracer using a Switch as the central device. All PCs communicate through the switch. Connectivity was verified using the ping command with all replies received successfully.

AIM

To create a Tree Topology using Cisco Packet Tracer and verify connectivity between the computers using the Ping command.

PROCEDURE**Step 1:** Open Cisco Packet Tracer

- Launch Cisco Packet Tracer and log in. A blank workspace appears.

Step 2: Add Network Devices

- From Network Devices → Switches, drag and place two Switch-2960 units: Switch0 (top / root) and Switch1 (bottom / leaf). These form the hierarchical tree structure.
- From End Devices, drag and place four PCs: PC0, PC1, PC2, PC3.

Step 3: Connect Switches Together (Trunk Link)

- Select Copper Straight-Through cable. Click on Switch0 FastEthernet0/1 → then click Switch1 FastEthernet0/1.
- This inter-switch link represents the 'trunk' of the tree. Allow a few seconds for the Spanning Tree Protocol (STP) to converge — the link light turns green.

Step 4: Connect PCs to Their Respective Switches

- Connect PC0 → Switch0 FastEthernet0/2 using Copper Straight-Through cable.
- Connect PC1 → Switch0 FastEthernet0/3 using Copper Straight-Through cable.
- Connect PC2 → Switch1 FastEthernet0/2 using Copper Straight-Through cable.
- Connect PC3 → Switch1 FastEthernet0/3 using Copper Straight-Through cable.
- Verify all link lights are green.

Step 5: Assign IP Addresses to All PCs

- For each PC: Double-click → Desktop → IP Configuration → Static.

Device	Connected To	IP Address	Subnet Mask
PC0	Switch0 (root)	192.168.1.1	255.255.255.0
PC1	Switch0 (root)	192.168.1.2	255.255.255.0
PC2	Switch1 (leaf)	192.168.1.3	255.255.255.0
PC3	Switch1 (leaf)	192.168.1.4	255.255.255.0

Step 6: Test Connectivity Within Same Switch (PC0 to PC1)

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.1.2
```

- This tests communication between two PCs on the same switch (Switch0). Should receive 4 replies.

Step 7: Test Connectivity Across Switches (PC0 to PC2)

- From PC0 Command Prompt, type:

```
ping 192.168.1.3
```

- This packet travels: PC0 → Switch0 → inter-switch link → Switch1 → PC2. This tests the hierarchical forwarding of the tree topology. Should receive 4 replies.

Step 8: Verify Complete Connectivity

- Test all remaining combinations from PC0:

```
ping 192.168.1.4
```

Step 9: Record Observations

Ping Attempt	From PC	To PC	Result
1	PC0	PC1	Reply received / Timed out
2	PC0	PC2	Reply received / Timed out
3	PC0	PC3	Reply received / Timed out
4	PC1	PC2	Reply received / Timed out

RESULT

Tree Topology was successfully created using two switches in a hierarchical arrangement. Connectivity was verified within the same switch level and across the switch hierarchy using the ping command.

AIM

To create a Mesh Topology using Cisco Packet Tracer and verify connectivity between the computers using the Ping command.

PROCEDURE**Step 1:** Open Cisco Packet Tracer

- Launch Cisco Packet Tracer. Log in and start with a blank workspace.

Step 2: Add PCs to the Workspace

- From End Devices, drag and place four PCs: PC0 (top-left), PC1 (top-right), PC2 (bottom-left), PC3 (bottom-right).
- Arrange them in a square/diamond pattern so the cross-over cable connections will be visually clear.

Step 3: Understand Mesh Topology Requirements

- In a Full Mesh Topology, every device connects directly to every other device. For 4 PCs, this means: PC0-PC1, PC0-PC2, PC0-PC3, PC1-PC2, PC1-PC3, PC2-PC3. Total of 6 direct connections.
- Since PCs connect directly to each other (peer-to-peer), Copper Cross-Over cables are used (same device type to same device type).

Step 4: Connect All PCs in Mesh Pattern

- Select Copper Cross-Over cable. Make the following connections one at a time:
- PC0 FastEthernet0 → PC1 FastEthernet0
- PC0 FastEthernet1 → PC2 FastEthernet0
- PC0 FastEthernet2 → PC3 FastEthernet0
- PC1 FastEthernet1 → PC2 FastEthernet1
- PC1 FastEthernet2 → PC3 FastEthernet1
- PC2 FastEthernet2 → PC3 FastEthernet2
- Note: Each PC needs multiple network interface cards (NICs). If a PC only has one NIC (FastEthernet0), you may need to add additional NICs. Double-click PC → Physical tab → drag a PT-HOST-NM-1CFE module into the NIC slot while the device is powered off.

Step 5: Assign IP Addresses

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Step 6: Test Connectivity

- Double-click PC0 → Desktop → Command Prompt. Test connectivity to all other PCs:

```
ping 192.168.1.2
ping 192.168.1.3
```

```
ping 192.168.1.4
```

- Each ping should return 4 replies. Multiple redundant paths exist between any two devices.

Step 7: Observe Redundancy

- In a Mesh Topology, if one direct link fails, data can reach the destination via an alternative path through another PC. This provides maximum redundancy and fault tolerance.

Step 8: Record Observations

Ping Attempt	From PC	To PC	Result
1	PC0	PC1	Reply received / Timed out
2	PC0	PC2	Reply received / Timed out
3	PC0	PC3	Reply received / Timed out
4	PC1	PC2	Reply received / Timed out

RESULT

Mesh Topology was successfully created in Cisco Packet Tracer with direct connections between all four PCs. All 6 peer-to-peer links were established and connectivity was verified using the ping command.

AIM

To create a Ring Topology using Cisco Packet Tracer and verify connectivity between the computers using the Ping command.

PROCEDURE**Step 1:** Open Cisco Packet Tracer

- Launch Cisco Packet Tracer and log in. Start with a blank workspace.

Step 2: Add PCs to the Workspace

- From End Devices, drag and place four PCs: PC0 (top), PC1 (right), PC2 (bottom), PC3 (left). Arrange them in a circle/ring pattern to visually represent the ring structure.

Step 3: Understand Ring Topology Connections

- In a Ring Topology, each device connects to exactly two other devices — its left neighbor and its right neighbor — forming a closed loop. Each PC needs two NICs.
- The connections form the ring: PC0 → PC1 → PC2 → PC3 → back to PC0.

Step 4: Add Additional NICs to Each PC

- Each PC in a ring needs two NIC ports. Double-click PC0 → Physical tab.
- Click the Power button to turn the PC off (red button turns grey).
- Drag a PT-HOST-NM-1CFE NIC module from the module panel into an empty slot.
- Click the Power button again to turn the PC on. Repeat for PC1, PC2, and PC3.

Step 5: Connect PCs in Ring Pattern

- Select Copper Cross-Over cable. Make the following connections:
- PC0 FastEthernet0 → PC1 FastEthernet0 (PC0 right side to PC1 left side)
- PC1 FastEthernet1 → PC2 FastEthernet0 (PC1 right side to PC2 left side)
- PC2 FastEthernet1 → PC3 FastEthernet0 (PC2 right side to PC3 left side)
- PC3 FastEthernet1 → PC0 FastEthernet1 (PC3 right side back to PC0 left side — closing the ring)
- Verify all 4 links show green. The ring is now complete.

Step 6: Assign IP Addresses

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

Step 7: Test Connectivity

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.1.2
```

```
ping 192.168.1.3
```

```
ping 192.168.1.4
```

- Data travels around the ring to reach the destination. Replies confirm successful ring communication.

Step 8: Note the Single Point of Failure

- A key limitation of Ring Topology: if any single link or PC fails, the entire ring is broken and communication stops. This is unlike Mesh Topology where redundant paths exist.

Step 9: Record Observations

Ping Attempt	From PC	To PC	Result
1	PC0	PC1	Reply received / Timed out
2	PC0	PC2	Reply received / Timed out
3	PC0	PC3	Reply received / Timed out
4	PC1	PC2	Reply received / Timed out

RESULT

Ring Topology was successfully created in Cisco Packet Tracer. All four PCs were connected in a closed ring using Copper Cross-Over cables. Connectivity was verified using the ping command.

AIM

To create a Hybrid Topology (combination of Star and Bus) using Cisco Packet Tracer and verify connectivity between the computers using the Ping command.

PROCEDURE**Step 1:** Open Cisco Packet Tracer

- Launch Cisco Packet Tracer. Log in and open a blank workspace.

Step 2: Plan the Hybrid Topology

- A Hybrid Topology combines two or more topologies. In this experiment, we create a combination of Star Topology (PCs connected to switches) linked together in a Bus-like manner (switches connected in series).
- Layout: Switch0 (left segment): PC0 and PC1 connect to Switch0. Switch1 (right segment): PC2 and PC3 connect to Switch1. Switch0 and Switch1 are connected to each other, forming the hybrid.

Step 3: Add Devices

- From Network Devices, drag and place two Switch-2960 units: Switch0 and Switch1.
- From End Devices, drag and place four PCs: PC0, PC1 near Switch0 on the left; PC2, PC3 near Switch1 on the right.

Step 4: Connect PCs to Their Switches (Star Portions)

- Using Copper Straight-Through cable:
- Connect PC0 FastEthernet0 → Switch0 FastEthernet0/1
- Connect PC1 FastEthernet0 → Switch0 FastEthernet0/2
- Connect PC2 FastEthernet0 → Switch1 FastEthernet0/1
- Connect PC3 FastEthernet0 → Switch1 FastEthernet0/2

Step 5: Connect the Two Switches Together (Hybrid Link)

- Using Copper Straight-Through cable:
- Connect Switch0 FastEthernet0/24 → Switch1 FastEthernet0/24
- Wait for the STP to converge — link lights turn green (may take a few seconds).
- This inter-switch link connects the two Star segments, creating the Hybrid Topology.

Step 6: Assign IP Addresses

Device	Segment	IP Address	Subnet Mask
PC0	Switch0 (Left Star)	192.168.1.1	255.255.255.0
PC1	Switch0 (Left Star)	192.168.1.2	255.255.255.0
PC2	Switch1 (Right Star)	192.168.1.3	255.255.255.0
PC3	Switch1 (Right Star)	192.168.1.4	255.255.255.0

Step 7: Test Connectivity Within Same Segment

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.1.2
```

- This tests communication within the left Star segment (PC0 → Switch0 → PC1). Should succeed.

Step 8: Test Connectivity Across Segments

- From PC0 Command Prompt, type:

```
ping 192.168.1.3
```

```
ping 192.168.1.4
```

- These packets travel: PC0 → Switch0 → inter-switch link → Switch1 → PC2/PC3. Should receive 4 replies each.

Step 9: Record Observations

Ping Attempt	From PC	To PC	Result
1	PC0	PC1	Reply received / Timed out
2	PC0	PC2	Reply received / Timed out
3	PC0	PC3	Reply received / Timed out
4	PC1	PC2	Reply received / Timed out

RESULT

Hybrid Topology combining two Star segments connected via an inter-switch link was successfully created. Connectivity within each segment and across segments was verified using the ping command.

AIM

To configure a DHCP server device in Cisco Packet Tracer to automatically assign IP addresses to client computers.

PROCEDURE**Step 1:** Open Cisco Packet Tracer and Build the Network

- Launch Cisco Packet Tracer. Place the following devices on the workspace: 1 Switch-2960, 1 Server (from End Devices → Servers), 3 PCs (PC0, PC1, PC2).
- Connect all devices to the switch using Copper Straight-Through cables: Server → Switch FastEthernet0/1, PC0 → Switch FastEthernet0/2, PC1 → Switch FastEthernet0/3, PC2 → Switch FastEthernet0/4.
- Verify all link lights are green.

Step 2: Assign a Static IP Address to the Server

- The DHCP server itself must have a static (manually assigned) IP address — it cannot get one from DHCP.
- Double-click the Server → Desktop tab → IP Configuration → select Static.
- Enter: IP Address: 192.168.1.1 | Subnet Mask: 255.255.255.0 | Default Gateway: 192.168.1.1
- Close the IP Configuration window.

Step 3: Open the DHCP Service on the Server

- Double-click the Server → Services tab → click DHCP in the left panel.
- The DHCP configuration panel opens. Toggle the DHCP service ON (switch turns green).

Step 4: Delete the Default Pool and Create a New Pool

- You will see a default pool named 'serverPool'. To avoid conflicts, first delete it by selecting it and clicking Remove.
- Now create a new pool by filling in the fields: Pool Name: Network1 | Default Gateway: 192.168.1.1 | DNS Server: 192.168.1.1 | Starting IP Address: 192.168.1.10 | Subnet Mask: 255.255.255.0 | Maximum Number of Users: 50.
- Click Add to save the new DHCP pool. 'Network1' appears in the pool list.

Step 5: Configure PC0 to Use DHCP

- Double-click PC0 → Desktop → IP Configuration.
- Select the DHCP radio button (instead of Static). The PC immediately sends a DHCP Discover broadcast.
- Within a few seconds, the PC receives an IP address from the server. The fields auto-populate: IP Address: 192.168.1.10 (or next available in pool), Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1.

Step 6: Configure PC1 and PC2 to Use DHCP

- Repeat Step 5 for PC1 and PC2. Each receives the next available IP from the pool: PC1 gets 192.168.1.11, PC2 gets 192.168.1.12.

Step 7: Verify DHCP Leases on the Server

- Double-click the Server → Services → DHCP.
- Scroll down to the DHCP Binding Table. It lists each PC's MAC address, assigned IP address, and lease expiry time. This confirms the server successfully assigned IPs.

Step 8: Verify Connectivity Between PCs

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.1.11
ping 192.168.1.12
```

- Both pings should receive replies, confirming the DHCP-assigned IPs are routable and connectivity works.

OBSERVATION — DHCP Address Assignment

Device	Assignment Method	IP Address Assigned	Subnet Mask	Gateway
Server	Static (manual)	192.168.1.1	255.255.255.0	192.168.1.1
PC0	DHCP (automatic)	192.168.1.10	255.255.255.0	192.168.1.1
PC1	DHCP (automatic)	192.168.1.11	255.255.255.0	192.168.1.1
PC2	DHCP (automatic)	192.168.1.12	255.255.255.0	192.168.1.1

RESULT

DHCP server was successfully configured on the Server device. Client PCs (PC0, PC1, PC2) automatically received IP addresses, subnet masks, and gateway information from the DHCP pool. Connectivity between all devices was verified using the ping command.

AIM

To configure a router as a DHCP server to automatically assign IP addresses to network devices in Cisco Packet Tracer.

PROCEDURE

Step 1: Build the Network Topology

- Open Cisco Packet Tracer. Place the following devices: 1 Router-2911, 1 Switch-2960, 3 PCs (PC0, PC1, PC2).
- Connect: Router FastEthernet0/0 → Switch FastEthernet0/1. PC0 → Switch FastEthernet0/2. PC1 → Switch FastEthernet0/3. PC2 → Switch FastEthernet0/4.
- All connections use Copper Straight-Through cables. Verify green link lights.

Step 2: Configure the Router Interface

- Double-click the Router → CLI tab. Press Enter to activate the CLI. Type:

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

- The FastEthernet0/0 interface is now the gateway for the connected network. The link light should turn green after 'no shutdown'.

Step 3: Exclude the Router's IP From the DHCP Pool

- The router's own IP (192.168.1.1) must be excluded so DHCP does not assign it to a client:

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.9
```

- This excludes 192.168.1.1 through 192.168.1.9 from the pool — leaving 192.168.1.10 onwards for clients.

Step 4: Create the DHCP Pool

```
Router(config)# ip dhcp pool NETWORK1
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# lease 7
Router(dhcp-config)# exit
Router(config)# exit
Router# write memory
```

- network — defines which IP range belongs to this pool. default-router — the gateway assigned to clients. dns-server — DNS server address for name resolution. lease 7 — lease duration in days (7 days). write memory — saves the configuration.

Step 5: Configure PCs to Obtain IP Automatically

- For each PC (PC0, PC1, PC2): Double-click the PC → Desktop → IP Configuration → select DHCP.

- Each PC sends a DHCP Discover. The router responds with an Offer, then Acknowledges the Request.
- Verify the PCs receive: IP: 192.168.1.10, 192.168.1.11, 192.168.1.12 respectively. Subnet Mask: 255.255.255.0. Gateway: 192.168.1.1.

Step 6: Verify DHCP Bindings on the Router

- On the Router CLI, type:

```
Router# show ip dhcp binding
```

- This displays all active DHCP leases — the IP address assigned to each MAC address and the lease expiry time.

Step 7: Verify DHCP Pool Statistics

- On the Router CLI, type:

```
Router# show ip dhcp pool
```

- This shows the pool name, network, available addresses, and count of addresses currently leased.

Step 8: Test Connectivity

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.1.11
```

```
ping 192.168.1.12
```

- Both should receive replies. Also ping the router:

```
ping 192.168.1.1
```

OBSERVATION — DHCP Bindings

Device	MAC Address	IP Assigned	Lease Expiry	Gateway
PC0	(auto-detected)	192.168.1.10	7 days	192.168.1.1
PC1	(auto-detected)	192.168.1.11	7 days	192.168.1.1
PC2	(auto-detected)	192.168.1.12	7 days	192.168.1.1

RESULT

The router was successfully configured as a DHCP server. All three client PCs automatically received IP addresses, subnet masks, gateways, and DNS server information from the router's DHCP pool. Connectivity was verified using the ping command.

AIM

To verify automatic IP assignment using DHCP and test connectivity between devices using the Ping command.

PROCEDURE

Step 1: Set Up the Network

- If continuing from Experiment 24 or 25, the DHCP server/router is already configured. Otherwise, set up: 1 DHCP Server (or Router configured as DHCP), 1 Switch, 3 PCs — all connected with Copper Straight-Through cables.

Step 2: Ensure PCs Are Set to DHCP Mode

- For each PC: Double-click → Desktop → IP Configuration → select the DHCP radio button.
- If the IP fields remain empty after a few seconds, click the DHCP button once more or wait up to 10 seconds for the DHCP process to complete.

Step 3: Verify IP Assignment on Each PC

- Double-click PC0 → Desktop → Command Prompt. Type:

```
ipconfig
```

- Record the IPv4 Address, Subnet Mask, and Default Gateway displayed. These should match the DHCP pool configuration (e.g., 192.168.1.10, 255.255.255.0, 192.168.1.1).
- Repeat for PC1 and PC2. Verify they received sequential IPs (192.168.1.11, 192.168.1.12).

Step 4: Test Connectivity from PC0 to PC1

- From PC0's Command Prompt, type:

```
ping 192.168.1.11
```

- Expected output: 4 lines of 'Reply from 192.168.1.11: bytes=32 time<1ms TTL=128'. This confirms PC0 can reach PC1 using the DHCP-assigned address.

Step 5: Test Connectivity from PC0 to PC2

- From PC0's Command Prompt, type:

```
ping 192.168.1.12
```

- Expected: 4 replies received.

Step 6: Test Connectivity from PC1 to PC2

- Double-click PC1 → Desktop → Command Prompt. Type:

```
ping 192.168.1.12
```

- Expected: 4 replies received.

Step 7: Verify IP Using ipconfig /all

- On any PC, type:

```
ipconfig /all
```

- This shows detailed configuration: Physical Address (MAC), IPv4 Address, Subnet Mask, Default Gateway, DHCP Server IP, DNS Server IP, and Lease Obtained/Expires times.

Step 8: Release and Renew IP Address (Test DHCP Re-assignment)

- On PC0, type:

```
ipconfig /release
```

- This releases the DHCP lease. The IP address returns to 0.0.0.0 temporarily.
- Then type:

```
ipconfig /renew
```

- PC0 performs the full DHCP DORA process and receives the same (or a new available) IP address from the server.

OBSERVATION

Device	DHCP IP Assigned	Ping to PC1	Ping to PC2	Status
PC0	192.168.1.10	Sent	Sent	4/4 Reply
PC1	192.168.1.11	Received	Sent	4/4 Reply
PC2	192.168.1.12	—	Received	4/4 Reply

RESULT

DHCP IP assignment was verified successfully. All PCs obtained automatic IP addresses and connectivity between them was confirmed using the ping command with all replies received.

AIM

To study and understand various Cisco IOS show commands used for monitoring, troubleshooting, and analyzing router status and network information.

PROCEDURE

Step 1: Open Cisco Packet Tracer and Set Up a Simple Network

- Place a Router-2911 and a PC. Connect them with a Copper Straight-Through cable.
- Configure the router interface: Double-click Router → CLI → type:

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# hostname ISPRouter
ISPRouter(config)# exit
```

Step 2: Enter Privileged EXEC Mode

- All show commands require Privileged EXEC mode (indicated by the # prompt). Type:

```
ISPRouter> enable
```

- The prompt changes to ISPRouter#. You are now in Privileged EXEC mode.

Step 3: show version — View IOS Version and Hardware Info

```
ISPRouter# show version
```

- Output includes: Cisco IOS version, System uptime, Router model and hardware details, Amount of RAM and Flash memory, Number and type of interfaces, Configuration register value. This is the first command to run when troubleshooting — it tells you exactly what hardware and software you are dealing with.

Step 4: show interfaces — View Interface Status and Statistics

```
ISPRouter# show interfaces
```

- Output for each interface includes: Interface name and line status (up/up, up/down, administratively down), Hardware address (MAC), Internet address (IP/mask), MTU (Maximum Transmission Unit) size, Input/output errors, Bandwidth and duplex settings. Use this to diagnose interface errors and connectivity issues.

Step 5: show ip interface brief — Quick Interface Status Summary

```
ISPRouter# show ip interface brief
```

- A concise table showing: Interface name, IP address assigned, OK? (Yes/No), Method (NVRAM, manual, DHCP), Status (up/down), Protocol (up/down). This is the most commonly used command for a quick health check of all interfaces. 'up/up' means fully operational. 'up/down' means physical link but no Layer 3 protocol. 'admin down' means intentionally shut down.

Step 6: show ip route — View the Routing Table

```
ISPRouter# show ip route
```

- Displays all routes the router knows about. Route codes: C = Connected (directly connected network), S = Static (manually configured route), R = RIP learned route, O = OSPF learned route, * = Candidate default route. Each entry shows: route prefix, next-hop IP, outgoing interface, and administrative distance/metric.

Step 7: show arp — View the ARP Table

```
ISPRouter# show arp
```

- Displays the router's ARP cache: IP addresses and their corresponding MAC addresses for recently-communicating devices. Useful for verifying that hosts on connected networks are reachable at Layer 2.

Step 8: show running-config — View Current Configuration

```
ISPRouter# show running-config
```

- Displays the entire current configuration stored in RAM. Shows: hostname, interface configurations, routing protocols, DHCP pools, access lists, passwords (encrypted), etc. This is the 'blueprint' of how the router is currently operating.

Step 9: show flash — View Flash Memory Contents

```
ISPRouter# show flash
```

- Lists the files stored in Flash memory, primarily the Cisco IOS image file. Shows: filename, file size, and total/free Flash space. Useful for verifying which IOS version is loaded and checking available space before upgrades.

Step 10: show protocols — View Layer 3 Protocol Status

```
ISPRouter# show protocols
```

- Shows which Layer 3 protocols (e.g., IP) are configured on each interface and the corresponding IP address. Useful for a quick audit of all configured IP addresses.

OBSERVATION — Show Command Summary

Command	Key Information Displayed	Common Use
show version	IOS version, RAM, Flash, model, uptime	Check hardware and software baseline
show interfaces	Line/protocol status, errors, bandwidth	Diagnose physical and data link issues
show ip interface brief	All interfaces, IP, status — one-line summary	Quick health check of all interfaces
show ip route	Routing table: C, S, R, O routes	Verify routing is working correctly
show arp	IP-to-MAC mappings in ARP cache	Verify Layer 2 connectivity to hosts
show running-config	Complete current configuration in RAM	Review and audit device configuration
show flash	Flash contents and IOS image	Verify IOS version and available space
show protocols	L3 protocol status per interface	Audit all assigned IP addresses

RESULT

Various Cisco IOS show commands were successfully executed and analyzed. These commands provide critical information for monitoring router status, verifying configurations, diagnosing connectivity issues, and auditing network devices.

AIM

To examine and analyze WAN connectivity using Cisco IOS show commands and understand the status of Frame Relay connections between branch and main routers.

PROCEDURE

Step 1: Set Up the Multi-Router WAN Topology

- Open Cisco Packet Tracer. Place three Routers: Branch1 (left), Branch2 (right), Main (center).
- Connect Branch1 Serial0/0/0 → Main Serial0/0/0. Connect Branch2 Serial0/0/1 → Main Serial0/0/1.
- On each serial link, one router acts as DCE (provides clock) — configure clock rate on DCE side.

Step 2: Configure Basic IP Addresses

- Configure the following on Main Router CLI:

```
Router> enable
Router# configure terminal
Router(config)# hostname MainRouter
MainRouter(config)# interface Serial0/0/0
MainRouter(config-if)# ip address 10.0.0.1 255.255.255.252
MainRouter(config-if)# clock rate 64000
MainRouter(config-if)# no shutdown
MainRouter(config-if)# exit
MainRouter(config)# interface Serial0/0/1
MainRouter(config-if)# ip address 10.0.0.5 255.255.255.252
MainRouter(config-if)# clock rate 64000
MainRouter(config-if)# no shutdown
MainRouter(config-if)# exit
```

- Configure Branch1: Serial0/0/0 → IP 10.0.0.2/30. Configure Branch2: Serial0/0/1 → IP 10.0.0.6/30.

Step 3: Examine Branch1 Router — show running-config

- Double-click Branch1 → CLI. Type:

```
Router# show running-config
```

- Verify: hostname, interface Serial0/0/0 IP address, encapsulation type (HDLC by default on serial links), and that 'no shutdown' is present.

Step 4: Examine Branch1 — show ip interface brief

```
Router# show ip interface brief
```

- Look for Serial0/0/0: Status must be 'up', Protocol must be 'up'. If status is 'up' but protocol is 'down', there is a Layer 2 mismatch (e.g., encapsulation type differs between the two ends).

Step 5: Examine WAN Link — show interfaces serial

```
Router# show interfaces serial 0/0/0
```

- Observe: Line protocol status (up/down), Encapsulation type (HDLC or Frame Relay), BER (Bit Error Rate), Input/Output errors, Bandwidth and delay values, Last input/output times. This command is essential for diagnosing WAN link issues.

Step 6: Examine Frame Relay Specific Commands (if using Frame Relay encapsulation)

- If the WAN uses Frame Relay encapsulation, these commands apply. First configure Frame Relay encapsulation:

```
Router(config-if)# encapsulation frame-relay
```

- Then examine Frame Relay:

```
Router# show frame-relay map
```

- Shows the mapping of DLCIs (Data Link Connection Identifiers) to remote IP addresses. Each DLCI is a virtual circuit identifier in Frame Relay.

```
Router# show frame-relay pvc
```

- Shows PVC (Permanent Virtual Circuit) status: DLCI number, PVC status (ACTIVE/INACTIVE/DELETED), input/output packet counts.

```
Router# show frame-relay lmi
```

- Shows LMI (Local Management Interface) statistics: LMI type (Cisco, ANSI, or Q.933a), number of status enquiries sent/received, and last status update time.

Step 7: Examine Main Router

- Double-click Main Router → CLI. Run the same set of commands:

```
MainRouter# show running-config
MainRouter# show ip interface brief
MainRouter# show interfaces serial 0/0/0
MainRouter# show interfaces serial 0/0/1
MainRouter# show ip route
```

Step 8: Verify End-to-End Connectivity

- On Branch1 CLI, ping Branch2:

```
Branch1# ping 10.0.0.6
```

- If replies are received, the WAN connection is functional.

OBSERVATION — WAN Show Commands

Command	Purpose	Expected Output
show running-config	View full router configuration	All interface IPs and encapsulation shown
show ip interface brief	Quick interface status	Serial0/0/0: up, up — both links operational
show interfaces serial	Detailed WAN interface statistics	Encapsulation HDLC/FR, 0 errors
show frame-relay map	DLCI-to-IP address mapping	DLCI 102 at ip 10.0.0.2, dynamic
show frame-relay pvc	PVC status and statistics	DLCI ACTIVE, input/output counts
show frame-relay lmi	LMI health statistics	LMI type Cisco, enquiries matched

RESULT

WAN connections were successfully examined using Cisco IOS show commands. Frame Relay DLCI mappings, PVC status, LMI statistics, and interface states were analyzed and verified. Successful end-to-end communication between Branch and Main routers was confirmed.

AIM

To demonstrate the functions performed by Distribution Layer devices, including inter-VLAN routing, packet filtering, and route summarization, using Cisco Packet Tracer.

PROCEDURE

Step 1: Understand the Three-Layer Network Architecture

- The Cisco hierarchical network model has three layers: Access Layer — connects end devices (PCs, printers) to the network via access switches. Distribution Layer — aggregates access layer switches, performs inter-VLAN routing, applies policy (ACLs), and summarizes routes. Core Layer — provides fast backbone connectivity between distribution layer devices.
- In this experiment, the Distribution Layer device (a Layer 3 switch or router) is the focus.

Step 2: Build the Three-Layer Topology

- Place the following in the workspace: 1 Core Router (Router-2911) — at the top. 2 Distribution Switches (Layer 3 Switch-3560 or Router acting as distribution) — middle level. 4 Access Switches (Switch-2960) — bottom level. 4 PCs — one connected to each access switch.
- Connect Core Router to both Distribution devices. Connect each Distribution device to two Access switches. Connect one PC to each Access switch.

Step 3: Configure VLANs on Access Switches

- On each Access Switch CLI, create VLANs for different departments:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
Switch(config)# interface FastEthernet0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
```

Step 4: Configure Trunk Ports Between Distribution and Access Switches

- On the Distribution switch uplink ports (connecting to access switches), configure trunk mode to carry all VLAN traffic:

```
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

Step 5: Configure Inter-VLAN Routing on the Distribution Layer

- The Distribution Layer performs inter-VLAN routing using Switched Virtual Interfaces (SVIs):

```
DistSwitch(config)# ip routing
```

```

DistSwitch(config)# interface vlan 10
DistSwitch(config-if)# ip address 192.168.10.1 255.255.255.0
DistSwitch(config-if)# no shutdown
DistSwitch(config-if)# exit
DistSwitch(config)# interface vlan 20
DistSwitch(config-if)# ip address 192.168.20.1 255.255.255.0
DistSwitch(config-if)# no shutdown
DistSwitch(config-if)# exit

```

Step 6: Configure EIGRP for Dynamic Routing (Distribution to Core)

- Configure EIGRP between the Distribution and Core layers for route exchange:

```

DistSwitch(config)# router eigrp 100
DistSwitch(config-router)# network 192.168.10.0 0.0.0.255
DistSwitch(config-router)# network 192.168.20.0 0.0.0.255
DistSwitch(config-router)# no auto-summary
DistSwitch(config-router)# exit

```

Step 7: Observe EIGRP Updates in Simulation Mode

- Switch to Simulation Mode. In the Event Filters, enable only EIGRP.
- Observe EIGRP Hello packets being exchanged between Distribution and Core devices — these maintain neighbor relationships and update routing tables.

Step 8: Test Connectivity in Realtime Mode

- Switch back to Realtime Mode. Double-click PC0 → Desktop → Command Prompt. Type:

```
ping 192.168.20.2
```

- A successful reply means traffic from VLAN 10 has been routed through the Distribution Layer to reach VLAN 20 — demonstrating inter-VLAN routing.
- Also test pings to PC2, PC3, and PC4.

Step 9: Apply an Access Control List (Packet Filtering) — Distribution Layer Function

- The Distribution Layer filters traffic between VLANs using ACLs. Apply a sample ACL:

```

DistSwitch(config)# ip access-list extended FILTER_POLICY
DistSwitch(config-ext-nacl)# permit ip 192.168.10.0 0.0.0.255 192.168.20.0
0.0.0.255
DistSwitch(config-ext-nacl)# deny ip any any
DistSwitch(config-ext-nacl)# exit
DistSwitch(config)# interface vlan 10
DistSwitch(config-if)# ip access-group FILTER_POLICY in
DistSwitch(config-if)# exit

```

OBSERVATION — Distribution Layer Functions

Function	Implementation	Verified By
Inter-VLAN Routing	SVI interfaces on Distribution Layer switch	Ping from VLAN 10 PC to VLAN 20 PC
Route Summarization	EIGRP no auto-summary with specific networks	show ip route on Core router
Packet Filtering (ACL)	Extended ACL applied on VLAN interface	Traffic between VLANs filtered as configured

EIGRP Routing Updates	EIGRP process 100 with network statements	EIGRP Hello packets in Simulation Mode
-----------------------	---	--

RESULT

Distribution Layer functions were successfully demonstrated including inter-VLAN routing using SVIs, packet filtering using ACLs, route summarization, and dynamic routing using EIGRP. EIGRP routing behavior was observed using Simulation Mode.

AIM

To study and execute basic network commands on a PC connected to a switch and analyze network connectivity and configuration.

PROCEDURE

Step 1: Set Up the Network

- Open Cisco Packet Tracer. Place 2 PCs (PC0 and PC1) and 1 Switch-2960.
- Connect PC0 → Switch FastEthernet0/1 and PC1 → Switch FastEthernet0/2 using Copper Straight-Through cables.
- Assign IP addresses: PC0: 192.168.1.1/255.255.255.0 | PC1: 192.168.1.2/255.255.255.0.
- Double-click PC0 → Desktop → Command Prompt to open the command line.

Step 2: ipconfig — Display Basic IP Configuration

```
ipconfig
```

- Output shows: IPv4 Address: 192.168.1.1, Subnet Mask: 255.255.255.0, Default Gateway: (blank if not set).
- This is the quickest way to verify what IP address a device has. Use it whenever you need to check the network configuration of a PC.

Step 3: ipconfig /all — Display Detailed IP Configuration

```
ipconfig /all
```

- Output includes: Host Name (computer name), Physical Address (MAC address, e.g., 00:60:47:AB:CD:EF), IPv4 Address, Subnet Mask, Default Gateway, DHCP Enabled (Yes/No), DHCP Server IP, DNS Servers, Lease Obtained and Lease Expires (if DHCP). This is the most informative network command — shows both Layer 2 (MAC) and Layer 3 (IP) information.

Step 4: ping — Test Connectivity

```
ping 192.168.1.2
```

- Tests whether PC1 is reachable from PC0. Output: 4 replies with round-trip times and TTL values (e.g., 'Reply from 192.168.1.2: bytes=32 time<1ms TTL=128'). 0% packet loss means full connectivity. 100% loss means unreachable.

Step 5: ping -t — Continuous Ping

```
ping -t 192.168.1.2
```

- Sends continuous ping packets without stopping. Useful for monitoring link stability over time. Press Ctrl + C to stop. Statistics show total sent, received, lost, and average round-trip time.

Step 6: ping with -n — Specify Packet Count

```
ping -n 10 192.168.1.2
```

- Sends exactly 10 ping packets instead of the default 4. Useful for more thorough connectivity testing over a longer period.

Step 7: tracert — Trace the Network Route

```
tracert 192.168.1.2
```

- Traces the path packets take from source to destination. For local network testing, only 1 hop appears. On internet connections, each router along the path is listed with its IP address and round-trip times. A * * * means a router did not respond (often due to ICMP filtering). Useful for identifying where packets are being dropped in the path.

Step 8: arp -a — Display ARP Table

```
arp -a
```

- Displays the PC's ARP cache — the table of IP-to-MAC address mappings learned from recent communication. Example output: 192.168.1.2 00-60-47-ab-cd-ef dynamic. 'dynamic' means it was learned automatically via ARP. 'static' means it was manually configured.

Step 9: netstat — Display Active Network Connections

```
netstat
```

- Shows all active TCP/UDP connections on the PC. Columns: Proto (TCP/UDP), Local Address (your IP:port), Foreign Address (remote IP:port), State (ESTABLISHED, TIME_WAIT, LISTEN). Useful for identifying active network sessions and listening services. In Cisco Packet Tracer this command shows simplified output.

Step 10: route print — Display Routing Table

```
route print
```

- Displays the PC's routing table. Entries include: Network Destination, Netmask, Gateway, Interface, Metric. Key entries: 0.0.0.0 entry = Default Gateway (all traffic goes here if no specific route). 127.0.0.0 = Loopback. Your IP's network = Directly connected.

Step 11: nslookup — Test DNS Resolution

```
nslookup google.com
```

- Queries the DNS server to resolve a domain name. Output: Server (DNS server IP), Name (queried domain), Address (resolved IP). Useful for verifying DNS configuration and troubleshooting name resolution issues. In Packet Tracer, a DNS server must be configured and running for this to work.

Step 12: ipconfig /release — Release DHCP Lease

```
ipconfig /release
```

- Releases the current DHCP-assigned IP address. The PC sends a DHCP Release message to the server. After release, IP becomes 0.0.0.0 and network access is lost until renewed.

Step 13: ipconfig /renew — Renew DHCP Lease

```
ipconfig /renew
```

- Requests a new IP address from the DHCP server (performs DORA process). Use this after /release or when the DHCP lease has expired. The PC should receive a new (or the same) IP address from the server.

OBSERVATION — Network Commands Summary

Command	Purpose	Key Output
ipconfig	Display basic IP settings	IP, Subnet Mask, Gateway

ipconfig /all	Display full network configuration	IP, MAC, DNS, DHCP details
ping <IP>	Test connectivity to a device	Reply / Timed Out — 4 packets
ping -t <IP>	Continuous connectivity test	Ongoing replies until Ctrl+C
tracert <IP>	Trace route to destination	Hop-by-hop path with latency
arp -a	View ARP cache	IP-to-MAC address mappings
netstat	View active connections	TCP/UDP connections and ports
route print	View PC routing table	Default gateway and network routes
nslookup <domain>	Test DNS name resolution	Resolved IP address of domain
ipconfig /release	Release DHCP IP address	IP set to 0.0.0.0
ipconfig /renew	Renew DHCP IP address	New IP received via DHCP

RESULT

All basic network commands were successfully executed on the PC connected to a switch in Cisco Packet Tracer. Commands covering IP configuration display, connectivity testing, route tracing, ARP table inspection, DNS resolution, and DHCP lease management were studied and analyzed.